

Do Recurrent RL Agents Discover a Unique Internal Clock?

Evidence Against a Canonical Temporal Code

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Summary

We examine how LSTM-based RL agents frame temporal information in their hidden states when no explicit clock is provided. Using CKA comparison with engineered encodings, linear probing, and PCA, we find that temporal encoding is training-dependent, non-unique across random seeds, and sensitive to temporal horizon. These findings challenge the implicit assumption that recurrent networks discover a single canonical temporal code, and suggest that the framing of time in learned representations is contingent on optimization path and task structure.

Contribution(s)

1. We show that temporal encoding in recurrent RL agents is not unique on complex environments—different training runs produce different internal clocks—but converges to a single code on simpler ones.
Context: Challenges the “canonical temporal code” assumption and links plurality to environment complexity.
2. We demonstrate that the representational geometry shifts with temporal horizon, from sinusoidal-like to scalar-like encoding.
Context: Links task structure to the learned framing of time, with implications for transfer and generalization.
3. We provide a simple, controlled methodology (CKA against engineered codes + multi-seed comparison) for testing uniqueness assumptions about learned representations.
Context: Applicable beyond time to other latent task variables.

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Abstract

1 Recurrent RL agents without an explicit clock must invent their own internal representa-
 2 tion of time. A natural assumption is that this representation is unique: any sufficiently
 3 trained agent converges to the same canonical temporal code. We show this assump-
 4 tion does not hold. LSTM agents trained on the same periodic task construct different
 5 temporal codes depending on their random initialization, despite identical behavioral
 6 performance. Using CKA, linear probing, and PCA on two periodic environments—
 7 a 2D gridworld (DoorGrid) and a 1D ring (Cyclic Corridor)—we ask three questions:
 8 (Q1) Is temporal structure learned or inherent to recurrence? (Q2) Is the emergent
 9 encoding unique across training runs? (Q3) How does the encoding depend on the
 10 temporal horizon? On DoorGrid, the learned encoding is *non-unique*: the highest-
 11 CKA engineered encoding varies across seeds, from sinusoidal positional encoding to
 12 phase-based encoding. On Cyclic Corridor, all seeds converge to the same encoding,
 13 suggesting that plurality depends on environment complexity. A controlled horizon ma-
 14 nipulation shows that the encoding is also *horizon-sensitive*: at short horizons, agents
 15 align with sinusoidal codes; at longer horizons, alignment shifts toward scalar encod-
 16 ing. These findings hold implications for how we interpret recurrent representations: if
 17 there is no single canonical temporal code, single-seed representational analyses may
 18 not generalize.

19 1 Introduction

20 RL researchers typically assume the hidden state of a recurrent network is a unified representation of
 21 the agent’s situation—including elapsed time. But when no clock signal is provided, the agent has to
 22 *construct its own framing* of the temporal dimension. This invites a question: given a well-defined
 23 POMDP, do trained agents converge to a *canonical* temporal code—a unique, reproducible internal
 24 clock? The answer is no: different training runs produce different temporal codes despite identical
 25 behavioral performance.

26 Prior work has established that time cells emerge in RNN-based RL agents, analogous to hippocam-
 27 pal time cells (MacDonald et al., 2011; Lin et al., 2023). In NLP, the form of positional encoding—
 28 sinusoidal, learned, or relative—has been a central design choice in the Transformer architecture
 29 (Vaswani et al., 2017). The *variability and contingency* of such representations in recurrent RL
 30 agents has received less attention.

31 We study three questions:

32(Q1) Does temporal structure emerge through training, or is it inherent to recurrent architectures?

33(Q2) Is the emergent encoding unique across training runs?

34(Q3) How does the encoding depend on the temporal horizon of the task?

We study these questions by training LSTM agents on DoorGrid, an environment with periodic dynamics, and comparing their hidden states against a library of engineered temporal encodings using Centered Kernel Alignment (CKA) (Kornblith et al., 2019), complemented by linear probing, PCA, and single-neuron phase selectivity analysis.

The temporal framing learned by LSTM agents is *non-unique*: agents with identical task performance align differently with simple engineered temporal codes, and the preferred alignment shifts with temporal horizon. This challenges how we interpret and compare recurrent representations across training runs—if there is no single canonical code, what exactly are we comparing?

2 Related Work

2.1 Temporal Representations in RL

Lin et al. (2023) showed that time cells emerge in RNN-based RL agents, extending the neuroscience observation that hippocampal neurons encode elapsed time (MacDonald et al., 2011). Hausknecht and Stone (2015) introduced DRQN, showing that recurrent connections enable RL agents to handle partial observability. Kapturowski et al. (2019) scaled this with R2D2 using LSTM-based agents on Atari. Jaderberg et al. (2017) showed that auxiliary temporal prediction tasks improve representation learning. Our work examines not just *whether* agents encode time, but whether this encoding has a unique canonical form.

2.2 Probing Neural Representations

Linear probing (Alain and Bengio, 2017) and CKA (Kornblith et al., 2019) are standard tools for analyzing representations. Hewitt and Manning (2019) emphasized that probe accuracy alone is insufficient—one must control for selectivity. Voita and Titov (2020) studied information-theoretic aspects of probing. CKA is used here as a descriptive similarity measure to a small library of candidate temporal codes, not as an identification procedure for the true latent code.

2.3 Multiple Solutions in Neural Networks

The existence of multiple loss landscape basins is well documented (Garipov et al., 2018; Li et al., 2018; Draxler et al., 2018). In neuroscience, Edelman and Gally (2001) formalized *degeneracy*—distinct structures performing the same function. We ask a question that sits at the intersection of these lines of work: do different optimization basins produce qualitatively different *temporal representations* for the same RL task?

3 Methods

3.1 Environment

DoorGrid is an 8×8 grid with a periodic door that alternates between open and closed. At $T = 4$, the door is open during phases 0–1 and closed during phases 2–3. The agent receives a one-hot encoding of its (x, y) position (64-dimensional); no explicit temporal features are provided. Actions are {up, down, left, right}. The reward is +1 for reaching the goal (top-right corner), −1 for stepping into a hazard (a cell that is terminal when the door is closed), and 0 otherwise. Episodes terminate upon reaching the goal, hitting a hazard, or exceeding a step limit. For horizon scaling, we vary $T \in \{4, 8, 16, 32\}$; the door is open during the first half of each period and closed during the second half.

This minimal environment isolates temporal encoding from complex spatial navigation: the periodic structure is the agent’s primary challenge.

Table 1: Trained vs. untrained LSTM on DoorGrid ($T = 4$). $R^2(t)$: timestep prediction. $\text{Acc}(\phi)$: phase classification. CKA Best: highest CKA with any engineered encoding. Return: best evaluation reward (20-episode mean). Trained row shows seed 42 as representative; all 5 seeds achieve $R^2 = 1.0$ and Return = 0.920. Untrained LSTM does not participate in the RL task, so Return is not applicable.

Condition	$R^2(t)$	$\text{Acc}(\phi)$	CKA Best	MI norm	Return
Trained (seed 42)	1.000	1.000	0.740	1.000	0.920
Untrained	0.304	0.298	0.153	0.082	N/A

Cyclic Corridor is a 1D ring of 32 cells with a periodic door at cell 15 (open during the first half of each period, closed during the second half). The goal is at cell 26; the agent starts at cell 0 and can move left or right (wrapping around). With $T = 16$, the state space is $32 \times 16 = 512$. This environment strips away all spatial complexity: there is no 2D navigation, no branching paths—only a periodic obstacle on a ring. If representational plurality depends on the complexity of the spatial task, Cyclic Corridor should produce more uniform encoding across seeds than DoorGrid.

3.2 Agents and Training

A single-layer LSTM ($d = 64$) with linear policy and value heads, trained with REINFORCE with baseline (learning rate 10^{-3} , entropy coefficient 0.01, discount $\gamma = 0.99$). An MLP baseline (2-layer, 64 units each) processes each observation independently, serving as a sanity check for the necessity of recurrence. All experiments use 5 random seeds {42, 123, 456, 789, 1024}.

3.3 Analysis Pipeline

After training, we run 200 evaluation episodes and collect hidden state vectors $\mathbf{h}_t \in \mathbb{R}^{64}$ along with timestep t and phase $\phi = t \bmod T$.

Linear Probe. Ridge regression from $\mathbf{h}_t$ to t (R^2 score) and logistic regression to ϕ (classification accuracy), with shuffled-target controls ($R^2_{\text{shuffled}} = 0.020$, confirming no leakage).

PCA. PC1 variance fraction and its correlation with timestep.

Mutual Information. Normalized mutual information $I(h_d; \phi)/H(\phi)$ between each hidden dimension and phase, averaged over dimensions, quantifying how much phase information is distributed across the representation.

CKA. Linear CKA (Kornblith et al., 2019) between $\mathbf{h}_t$ and engineered temporal encodings: raw scalar (t/T), sinusoidal PE ($\sin / \cos$ of $2\pi t/T$), phase $\sin/\cos$, one-hot phase, and remaining time $((T - t)/T)$. As reference baselines, CKA between engineered encodings themselves ranges from 0.38 to 0.54 at $T = 8$, and CKA between any encoding and shuffled hidden states is approximately 0.

Single-Neuron Phase Selectivity. For each hidden dimension d , we compute the mean activation per phase $\bar{h}_d(\phi)$ and a selectivity index $s_d = (\max_{\phi} \bar{h}_d - \min_{\phi} \bar{h}_d) / (\max_{\phi} \bar{h}_d + |\min_{\phi} \bar{h}_d| + \epsilon)$. Neurons with $s_d \approx 1$ are strongly phase-tuned. We compare the top- k phase-selective dimensions across seeds.

4 Results

4.1 Temporal Structure Emerges Through Training (Q1)

Table 1 speaks to Q1. Trained LSTMs show strong temporal encoding ($R^2 = 1.0$, CKA = 0.74, phase accuracy 1.0) while untrained LSTMs show CKA ≈ 0 and phase accuracy at chance (0.30 for

Table 2: CKA (mean $\pm$ std across 5 seeds) between DoorGrid ($T = 4$) hidden states and engineered encodings. High std relative to mean indicates seed-dependent encoding. Highest-CKA mean in **bold**.

Encoding	CKA (mean $\pm$ std)
Raw scalar (t/T)	0.40 ± 0.10
Sinusoidal PE	0.63 ± 0.09
Phase sin/cos	0.28 ± 0.21
One-hot phase	0.27 ± 0.16

Table 3: CKA (mean $\pm$ std across 5 seeds) on Cyclic Corridor ($T = 16$). All seeds converge to phase sin/cos as the best match—unlike DoorGrid, where the preferred encoding varies across seeds.

Encoding	CKA (mean $\pm$ std)
Raw scalar (t/T)	0.885 ± 0.010
Sinusoidal PE	0.760 ± 0.007
Phase sin/cos	0.896 ± 0.010
One-hot phase	0.471 ± 0.005

109 4 classes). Temporal structure is training-dependent, not an artifact of random recurrent dynamics.
 110 The MLP baseline (which lacks recurrence) fails to learn the task (Return = -2.0 across all 5 seeds,
 111 $R^2(t) \approx 0$), confirming that temporal tracking requires recurrent state.

112 4.2 The Temporal Framing Is Not Unique (Q2)

113 All 5 seeds achieve $R^2 = 1.0$, phase accuracy = 1.0, and Return = 0.920—functionally identical
 114 performance. Yet the CKA profiles differ substantially across seeds. Of 5 seeds, 4 show high-
 115 est CKA with sinusoidal PE (0.64–0.74), while seed 456 shows highest CKA with phase sin/cos
 116 (0.62) with sinusoidal PE scoring lower (0.46). The coefficient of variation across seeds is 0.15
 117 for CKA(sin_PE) and 0.76 for CKA(phase), indicating that the encoding geometry is sensitive to
 118 random initialization. Cross-seed CKA between the agents’ own hidden states (aligned by timestep)
 119 is uniformly high (> 0.96), so the overall representational geometry is similar across seeds. The
 120 plurality we identify is not in the raw hidden-state structure but in the *alignment with specific en-*
 121 *gineered codes*: the same population geometry can be more or less similar to different candidate
 122 temporal codes depending on initialization.

123 **Single-neuron evidence.** The top-8 phase-selective dimensions show low overlap across seeds
 124 (1.8/8, Jaccard = 0.14), but this is indistinguishable from chance given that 34 of 64 dimensions have
 125 near-maximal selectivity (hypergeometric expectation ≈ 1.88). We do not treat single-neuron over-
 126 lap as evidence for or against plurality; the population-level CKA differences provide the stronger
 127 signal.

128 4.3 Plurality Depends on Environment Complexity

129 To test whether representational plurality is specific to environments with spatial complexity, we
 130 repeat the multi-seed experiment on Cyclic Corridor. Table 3 shows a striking contrast with Door-
 131 Grid: all 5 seeds agree on phase sin/cos as the best-matching encoding, with low variance (CKA
 132 std = 0.010 vs. 0.21 for phase sin/cos on DoorGrid). The 1D ring eliminates spatial ambiguity, and
 133 with it, the seed-dependent encoding variability observed on DoorGrid. This suggests that represen-
 134 tational plurality emerges when the environment offers multiple viable encoding strategies—not as
 135 a universal property of recurrent networks.

Table 4: Representation similarity shifts with temporal horizon in DoorGrid. CKA (mean $\pm$ std, 5 seeds) with engineered temporal encodings. [†]1 seed failed at $T = 16$ and 2 at $T = 32$; stats computed on all 5. Converged-only: $T = 16$: CKA(raw)=0.59, sin_PE=0.65; $T = 32$: CKA(raw)=0.61, sin_PE=0.54.

T	CKA(raw)	CKA(sin_PE)	PC1-t	$R^2(t)$	Return
4	0.40 ± 0.10	0.63 ± 0.09	0.44 ± 0.10	1.000	0.920
8	0.56 ± 0.14	0.58 ± 0.08	0.46 ± 0.29	1.000	0.880
16	0.64 ± 0.15	0.60 ± 0.10	0.63 ± 0.37	1.000	$0.800^\dagger$
32	0.70 ± 0.22	0.50 ± 0.08	0.71 ± 0.28	1.000	$0.640^\dagger$

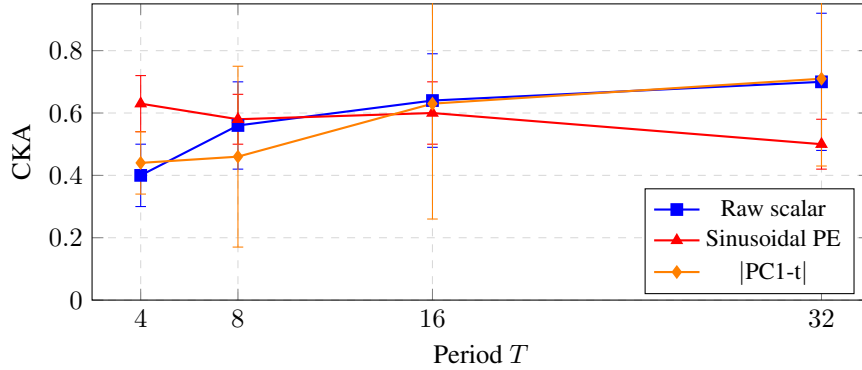


Figure 1: CKA with engineered encodings and |PC1-timestep correlation| as a function of period T (mean $\pm$ std across 5 seeds). Similarity to raw scalar encoding increases at longer horizons, while similarity to sinusoidal PE decreases.

136 4.4 The Framing Shifts with Temporal Horizon (Q3)

137 Table 4 and Figure 1 answer Q3. At $T = 4$, mean similarity to sinusoidal PE is highest (CKA
 138 = 0.63 vs. raw scalar = 0.40). At $T = 8$, the two are roughly equal (CKA $\approx$ 0.57). At $T \geq 16$,
 139 similarity to raw scalar exceeds sinusoidal PE. The |PC1-timestep correlation| also increases with
 140 period length (from 0.44 at $T = 4$ to 0.71 at $T = 32$), indicating that temporal information becomes
 141 more concentrated in a single dominant dimension at longer horizons.

142 The seed-level variability persists across all period lengths. At $T = 8$, seed 42 shows highest CKA
 143 with raw scalar (0.81) while seeds 123/456/789 show highest CKA with sinusoidal PE (0.48–0.58).
 144 At $T = 16$, seed 1024 matches raw scalar (0.84) while seed 789 matches remaining time (0.77).
 145 The framing of time is not an artifact of a single period length.

146 At $T = 32$, training becomes less stable: 2 of 5 seeds fail to converge, and task return drops to 0.640
 147 (see Appendix B for failed-seed training curves). The CKA shift at long horizons may partly reflect
 148 optimization difficulty rather than purely representational adaptation.

149 5 Discussion

150 LSTM-based RL agents do not converge to a single canonical temporal code—at least not in en-
 151 vironments with spatial complexity. On DoorGrid, the code depends on initialization, and which
 152 engineered encoding it resembles shifts with the task’s temporal horizon. On the simpler Cyclic
 153 Corridor, all seeds converge to the same encoding (phase sin/cos), suggesting that plurality is a
 154 property of the environment, not of the learning algorithm. These findings hold implications for
 155 how we interpret recurrent representations.

156 **Plurality is environment-dependent.** The assumption that recurrent networks provide a unified,
 157 well-defined temporal code is embedded in much of RL practice. Our results show this assumption
 158 is environment-dependent: on DoorGrid, the framing of time is contingent on optimization path,
 159 but on Cyclic Corridor, all seeds converge to the same code. Plurality arises when the environment
 160 offers multiple viable encoding strategies, not as a universal property of recurrent learning.

161 **Interpretability and representation.** If temporal codes are non-unique, then “what the agent has
 162 learned” is not a well-defined mapping from task to internal code, but one of many functionally
 163 equivalent solutions. This matches [Edelman and Gally \(2001\)](#)’s concept of degeneracy—structurally
 164 different elements performing the same function—and suggests that single-seed interpretability
 165 analyses may be misleading. A study using only seed 42 would conclude “LSTMs encode time
 166 via sinusoidal-like codes,” while seed 456 would suggest phase-based codes. Neither generalizes.

167 **Limitations.** Two caveats: first, our experiments use two environments, one algorithm (REIN-
 168 FORCE), and one architecture (single-layer LSTM). Second, CKA measures linear similarity and
 169 may miss nonlinear encoding strategies. Whether representational plurality persists across architec-
 170 tures and in more complex domains remains to be tested.

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217 A Single-Neuron Phase Selectivity Analysis

Table 5: Top-8 time cell dimensions and preferred phases per seed (DoorGrid, $T = 4$). Mean pairwise overlap: 1.8/8 dimensions (Jaccard = 0.14). Hypergeometric expectation given 34/64 selective dims: ≈ 1.88 .

Seed	Top-8 dims	Preferred phases
42	28, 43, 7, 44, 31, 32, 50, 39	2, 2, 0, 1, 3, 1, 3, 2
123	2, 40, 25, 49, 14, 4, 33, 36	1, 2, 3, 3, 3, 1, 3, 2
456	40, 35, 44, 39, 36, 22, 26, 43	3, 2, 0, 0, 3, 2, 3, 2
789	25, 51, 10, 15, 29, 18, 24, 38	1, 1, 1, 3, 3, 3, 3, 3
1024	43, 10, 25, 11, 39, 4, 36, 40	3, 1, 2, 3, 2, 1, 3, 2

- 218 Hidden-unit indices are not aligned across independently trained models, so low overlap in unit
 219 indices is partially expected under permutation invariance. With 34 of 64 dimensions showing
 220 high phase selectivity, the observed overlap (1.8/8) is close to the hypergeometric chance level
 221 (≈ 1.88), so this analysis does not provide evidence for or against representational uniqueness. The
 222 population-level CKA differences provide the stronger signal: seed 42 has $\text{CKA}(\sin_PE) = 0.74$
 223 while seed 456 has $\text{CKA}(\text{phase}) = 0.62$ with $\text{CKA}(\sin_PE) = 0.46$, indicating genuinely different
 224 encoding geometries.

225 B Failed Seeds at Long Horizons

- 226 At $T = 16$, seed 1024 fails to converge (best eval reward = -2.0). At $T = 32$, seeds 456 and
 227 789 fail to converge (best eval reward = -2.0). The reported CKA statistics in Table 4 include all
 228 5 seeds, as even non-converged agents contain representational structure worth characterizing. At
 229 $T = 32$, excluding the 2 failed seeds: $\text{CKA}(\text{raw}) = 0.61 \pm 0.28$ (vs. 0.70 ± 0.22 with all 5), and
 230 $\text{CKA}(\sin_PE) = 0.54 \pm 0.10$ (vs. 0.50 ± 0.08). The same pattern holds—raw scalar still beats sin
 231 PE.